

Course Work Project Description and Rubric

Semester	202410	Division	CIS
Assessment title in Syllabus	Project	Program	BSCIT

Course Code	CIS2003		
Course Title	Statistics and Probability		
Course Instructor	CLO1, CLO3, CLO4	CRN	
Assessment Weight	100	Duration & Deadlines	6 Weeks
Week/Date of Assessment	Report in Week 14, Oral in Week 15	% of Final Grade	40%

For Group Work submissions an additional individual assessment will be conducted.

Grades for the students in one group will vary based on the individual performance in the additional assessment.

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقاً لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبين وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض إلى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

Student (s) Signature: _____

Student Name(s):	Maryam Ali			
Student HCT ID(s):	H00535836			

For Examiner's Use Only

Section No.	Report	Oral Communication Follow-up Questions	Total	%
Marks Allocated	50	50	100	100
Marks Obtained				

Chatbot (e.g. ChatGPT) statement

This submission does not allow you to use any AI tool to complete. Any breach of this requirement will be treated as a plagiarism attempt and shall be subject to the HCT Academic Integrity Policy.

Data collection statement

This submission requires the group members to collect data utilizing different techniques like online forms, open-access websites, company reports...etc. It is essential for the group members to be cautious with the data they collect and protect private information. This includes being mindful of the types of information they collect, who they share it with, and how they store it.

1. This assessment has **11** pages including the cover page.
2. Students are allowed to use all available resources.
3. The group project report will be submitted on Blackboard by only ONE member of each team.

PROJECT OBJECTIVES

This course is a project-intensive course, designed to provide students with hands-on experience in the following key areas:

- Formulating hypotheses statements for a chosen research problem.
- **Planning and executing** the data collection process using authentic data sources.
- Producing a **descriptive statistics report** on a chosen sample data set.
- **Estimating population parameters** using confidence intervals.
- Conducting **hypothesis testing** to verify or dispute claims based on the collected data.

CLOs Covered in the Project:

1. **CLO1: Apply the key concepts of descriptive statistics**, including data summarization and data visualization, to practical problems.
2. **CLO3: Compute probabilities and expectations** of sampling and continuous probability distributions.
3. **CLO4: Use basic inferential statistical methods** to generalize from sample data using estimation techniques and hypothesis testing methods.

Competencies Assessed:

- **Descriptive Statistics (CLO1):** Competency in summarizing data using measures such as mean, median, mode, standard deviation, and producing visualizations (e.g., histograms, boxplots).
- **Sampling and Probability (CLO3):** Proficiency in applying sampling methods and computing probabilities related to continuous data distributions.
- **Inferential Statistics (CLO4):** Ability to apply estimation techniques (e.g., confidence intervals) and hypothesis testing to draw conclusions from sample data.

Employability Skills Developed:

1. **Data Analysis and Interpretation:**
 - Students will develop the ability to analyze real-world data sets, summarize key insights using descriptive statistics, and draw conclusions through hypothesis testing. These skills are critical for data analysts, statisticians, and business intelligence roles.
2. **Problem Solving:**
 - Students will enhance their problem-solving skills by applying statistical methods to practical, real-world problems. This process will include formulating research questions, choosing relevant data, and utilizing suitable analytical tools to address complex issues.
3. **Critical Thinking:**
 - Students will learn to critically evaluate data, check for normality, and test the validity of assumptions through hypothesis testing. This is an essential skill for roles in research, consulting, and decision-making.
4. **Collaboration and Teamwork:**
 - As the project is conducted in groups, students will gain experience in collaborative work, learning how to effectively divide tasks, communicate findings, and contribute to group efforts, skills vital for any modern workplace.

5. Communication and Presentation Skills:

- The individual oral defense and project report will help students develop their ability to communicate complex ideas clearly and present statistical findings effectively, which are key skills for professional presentations and client communications.

6. Technical Skills:

- Students will gain proficiency in Python (or other statistical tools) for data analysis, giving them hands-on experience with industry-relevant technology and analytical software used by data professionals.

PROJECT DESCRIPTION

Part A: Group work (50%)

This is a group project with a group size of 2 or 3 members. Each group will work on a single column of numerical data that represents a **continuous variable (a variable representing data that can be measured such as weight, height, length, temperature, speed, voltage...etc.)**

The following are the tasks to be completed in the project:

1. Use descriptive statistics to summarize the sample data collected. (Section 1)
2. Check for the normality of this data. (Section 2)
3. Estimate the population parameter (μ) at different Confidence Levels. (Section 3)
4. Perform hypothesis testing about the 'mean' to verify or dispute the claim in the initial research question (hypothesis). (Section 4)

Further requirements of the project are:

- Python code should be used as and when required for the tasks.
- An appropriately formatted MS Word report describing all the project tasks should be submitted at the end of the project. Do not copy the Project Tasks/Questions in this document into your report. Use proper headings and subheadings to organize the report.

Please note that SafeAssign has been enabled in this project submission. Students are allowed to view the SafeAssign originality report for their attempt.

Part B: Individual Oral Defense and Follow-up Questions (50%)

Each student will be assessed after the submission of the group project report through an oral defense/viva. Students are also expected to report with evidence on how the team worked together and the individual's contributions. This will contribute 50% of their project grade.

PROJECT TASKS/QUESTIONS

Section I: Data Collection and Descriptive Statistics [CLO1, CLO4]

(12 Marks)

- a) Each group will determine a **research problem** about population **mean** that requires collecting primary or secondary data for drawing conclusions based on hypothesis testing of single sample mean. An example could be 'Whether the average time it takes for HCT students to arrive to school is less than 15 minutes?' [CLO4] (1 Mark)

Our dataset is about top Instagram popular influencers from 3 different continent Asia, America, and Europe. In this project we will apply all the statistical techniques to analysis the data and report our findings. The project is divided into four sections and each section analyzes the data according to the instruction given.

- First section is about **Data Collection and Descriptive Statistics.**
- Second section is about **Checking the Normality of Population.**
- Third section is about **Estimation of Population Mean.**
- Fourth section is about **Hypothesis testing.**

We used the mean of the Instagram subscriber per 1000 Instagram influencers which was provided as the measure of central location, because mean is the common measure of central location.

Our research question is “Is the average number of followers for accounts in this dataset equal to 5 million?”

- b) Name an appropriate sampling method for data collection and explain why it is the best choice for your research problem. [Note: To produce meaningful interpretations, the data size should be large. Though a data sample of 30 is considered large statistically, collecting more than 30 data values is advised]. [CLO3]

Considering that our data consist of very large data with a population of 1000 influencers, We chose the **Systematic Random Sampling** method to implement since our data is organized in order starting from Europe, America, and then ending it with Asia.

- c) **Provide clear steps of how you implement the sampling method** if you gathered primary data. If your group used secondary data, describe the rationale for selecting a particular data source by explain its authenticity, and ensure that the data is appropriate for statistical analysis” [CLO3]

Our data is arranged in order by continent starting with Europe, America, and then Asia.

We have collected our sample by:-

- 1- Dividing the 1000 influencers (each influencer has number of followers) by 30 (wanted sample) meaning that we will get 30 groups each group consist of 33.3 items(influencers) then rounding it down to 33 items.
 $(1000/30) = 33.3 \rightarrow 33$
- 2- From each group we will select the first item (influencer) and then collect these 30 items(influencers) as our sample.

In conclusion, we will get 33 influencers as our sample from the 30 groups.

Sample table

1	cristiano	Cristiano f	https://ww Sports witi	400.1	Europe	50:24.8
34	uarmyhop	jhope	https://www.instagram	28.5	Europe	50:24.8
67	handemi	Hande Erc	https://www.Cinema A	26.7	Europe	50:24.8
100	jenniferan	Jennifer Ar	https://ww Cinema A	39.6	Europe	50:24.8
133	hazardede	Eden Haza	https://ww Sports witi	26.9	Europe	50:24.8
166	thedevera	Vijay Deve	https://ww Cinema A	14.6	Europe	50:24.8
199	hardikpan	Hardik Hin	https://ww Sports witi	20.2	Europe	50:24.8
232	ashishcha	Ashish Ch	https://ww Cinema A	12.5	Europe	50:24.8
265	nicki.nicole		https://ww Music	11.2	Europe	50:24.8
298	jhope_big	J-HOPE (i	https://ww Lifestyle	8.5	America	50:24.8
331	wow_kims	á,Éá...µá†	https://ww Music Cir	11.9	America	50:24.8
364	sanchooo	Jadon San	https://ww Sports witi	8.1	America	50:24.8
397	daquan	Daquan	https://ww Humor Fu	16.7	America	50:24.8
430	imouniroy	mon	https://ww Cinema A	21.8	America	50:24.8
463	floydmayw	Floyd May	https://ww Fitness G	27.8	America	50:24.8
496	nadin_seri	Nadin Ser	https://ww Family Lif	9.9	America	50:24.8
529	corinnako	pouty girl	https://ww Beauty	6.3	America	50:24.8
562	quavohun	QuavoHur	https://ww Music	20.7	America	50:24.8
595	thenewcla	Iggy Azale	https://ww Music	15.7	America	50:24.8
628	tiffanyyoui	tiffany you	https://ww Modeling	11.4	America	50:24.8
661	belindapo	Beli	https://ww Music	14.6	America	50:24.8
694	pokimane	pokimane	https://ww Lifestyle S	6	America	50:24.8
727	theshilpas	Shilpa She	https://ww Cinema A	24.2	Asia	50:24.8
760	thegirljt	JTðY;†	https://ww Music	5	Asia	50:24.8
793	miketyson	Mike Tysor	https://ww Fitness G	16.3	Asia	50:24.8
826	actor_jg	ðY™ë°°i§°	https://ww Cinema A	6	Asia	50:24.8
859	brycehall	Bryce Hall	https://ww Music	7.6	Asia	50:24.8
892	from_jilee	jung jae Le	https://www.instagram	5.3	Asia	50:24.8
925	becaa	bai Rebeca	ðYhttps://www.instagram	7.5	Asia	50:24.8

d) Descriptive Statistics -Measures of central tendency and dispersion. [CLO1] (4 Marks)

- i. Briefly explain the type/nature of the data that will be collected for the research

problem including the reason why the data is continuous. (1 mark)

Our data are more likely to be continuous which means that the data can take any value within a certain range and not be confined to particular values. In this case, it is believed to be continuous because the number of followers can range from 0 to any positive integer.

- ii. Show the raw data you collected and calculate the measures of central tendency and dispersion. Round the calculated values appropriately. (2 marks)

Our sample data :

400.1	72.3	26.7	39.6	26.9	14.6
20.2	12.5	11.2	8.5	11.9	8.1
16.7	21.8	27.8	9.9	6.3	20.7
15.7	11.4	14.6	6	24.2	5
16.3	6	7.6	5.3	7.5	8.3

The measure of central tendency and dispersion(variation):

Measures of central tendency and variation	value
Mean ($\bar{x}$)	29.46
Median	13.55
Mode	6
Sample standard deviation (σ)	71.27
Sample variance (σ^2)	5079.5
Range(max - min)	400.1 – 5 = 395.1

Briefly write your interpretations about the calculated measures of central tendency and dispersion. (1 marks)

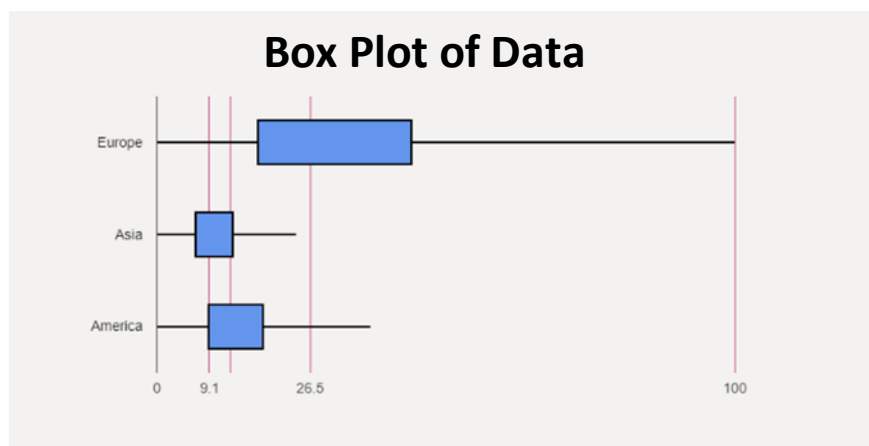
There is a huge variation within this dataset, with the mean at 29.46 and a standard deviation of 71.27. The fact that the median equals 13.55 and the mode equals 6 means this data is right-skewed, with a few large values pulling the mean upward. The high variance further shows the spread of the data points.

e) Descriptive Statistics- Graphical Representation [CLO1]

(4 Marks)

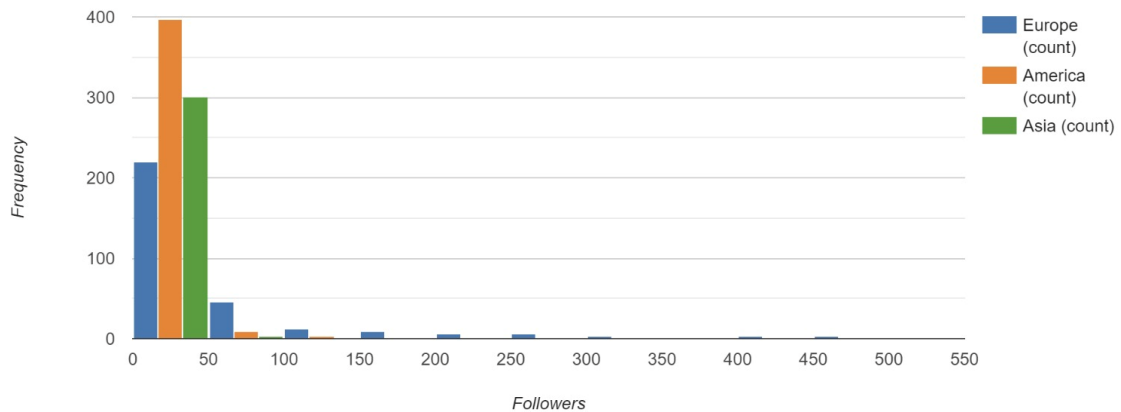
- i. Use software (Python) to describe the data graphically with a Boxplot and a Histogram. List the 5-number summary of the data, and frequency distribution. Include the Python codes in the report. (3 marks)

Groups	Europe	Asia	America
Minimum:	0	0	0
Q1:	17.25	6.4	8.75
Median:	26.5	9.1	12.6
Q3:	44.25	13.3	18.5
Maximum:	100	24.2	37



- Europe has a median of 26.5, and its median falls to the left of the center, the distribution is right skewed which is Positively skewed.
- The median of Asia appears to be 9.1, which falls on the left of the center of the box, the distribution is positively skewed.
- The median of America is 12.6 the box plot is slightly skewed to the right, indicating a slightly longer tail on the right side.

Followers across three regions



Our histogram provides the distribution of the followers across three regions: Europe, America, and Asia. The distribution of followers in all three regions is right-skewed, meaning that majority of the accounts have a low number of followers, but a very small percentage of the accounts have significantly high numbers of followers.

- Europe has the largest number of accounts with the lower number of followers.
- America has the highest number of accounts with a high following count.
- Asia is relatively balanced, on the other hand.

-----END OF SECTION-----

SECTION II: Checking the Normality of Population [CLO3]

(5 Marks)

Read pages 155, 165-166, and 326-328 of the textbook (provided in BBL).

- a) Using the histogram constructed in Section I sub-section e, calculate Pearson's coefficient of skewness and check for outliers. [CLO3] (4 Marks)

Calculating the Pearson's coefficient :

The formula that we are going to use is :

$$PC = \frac{3(\bar{X} - \text{median})}{s}$$

$$PC = \frac{3(24.84 - 14.4)}{38.70} = 0.809 \approx 0.81$$

$\bar{x}$ = Mean = 24.84

S = population Standard Deviation = 38.70

Median = 14.4

In our case, the Pearson's coefficient is 0.81 which is not greater than +1 or less than -1, so it can be concluded that the distribution is not significantly skewed. But we can clearly see that it's positively skewed in the histogram. Which the PC insures us that it's positively skewed ($0 > 0.81$).

Checking the Outliers:

Five number Summary	Values
Minimum	2.8
Q1	9
Q2 = Median	14.4
Q3	25.65
Maximum	469.6

Step1: find IQR

$$\text{IQR} = Q3 - Q1 = 25.65 - 9$$

$$\text{IQR} = 16.65$$

Step2: Multiply IQR by 1.5

$$16.65 * 1.5 = 24.975$$

Step3: Subtract the value obtained in Step 2 from Q1 and add it to Q3.

Lower limit = $Q1 - 24.975$	Upper limit = $Q3 + 24.975$
= $9 - 24.975$	= $25.65 + 24.975$
= - 15.975	= 50.625

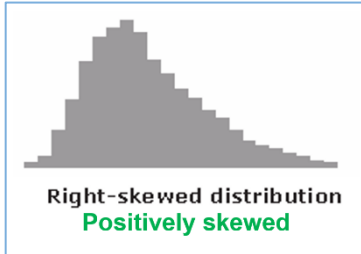
Step4 : Check the data set for values less than (- 15.975) or grater than (50.625)

Outliers

469.6, 400.1, 308.8, 306.3, 295.8, 295.8, 294.1, 284.9, 237.2, 219.8, 219.4, 217.8, 206.6, 199, 194.9, 182.6, 174, 169.8, 162.2, 160.2, 151.9, 136.6, 127.8, 125.8, 123.1, 120.9, 114.4, 111.4, 109.8, 105.1, 103.9, 102.6, 100, 87.9, 78.7, 78.2, 74, 73.9, 73.6, 73.1, 72.3, 71.6, 71.1, 69.2, 68.9, 67.7, 67.6, 66.5, 65.8, 65.7, 65.3, 64.2, 63.8, 63.3, 62.3, 61.6, 61.3, 60.9, 60.9, 60, 59.5, 59.3, 59.1, 58.9, 58, 58, 57.9, 57.1, 56.9, 56.1, 56, 55.1, 54.7, 53.7, 53.5, 52.3, 52.3, 52.1, 51.9, 51.5, 51.5, 51.4, 51.4, 51.4

We found out that we have 84 values. Since all values are greater than the upper limit (50.625). Also, considering that our data are the followers of the influencers (people), it would be impossible to find an influencer with a negative number of followers, which means that there is no lower outlier.

- b) Make a final comment (using the values/findings from the above step) about the normality of the data collected- whether it is near normal or clearly normal. [CLO3] (1 Mark)



A Pearson correlation coefficient of 0.81 indicates that the data are positively skewed and there is a strong positive correlation between the two variables. This suggests that as one variable increases, the other variable tends to increase as well. However, the presence of the outlier can significantly influence the correlation coefficient, and the outliers help to get a more accurate relation between variables.

-----END OF SECTION-----

SECTION III: Estimation of Population Mean [CLO4]

(16 Marks)

- a) What is the point-estimate for the population mean?

(1 Mark)

The point estimate of the population mean μ is the sample mean $\bar{X}$.

In this case its $\bar{X} = 29.45666 = 29.45$

- b) Construct confidence intervals for the population mean using 3 different confidence levels (90 %, 95%, and 99%) [Note: these calculations should be **done by hand** and not using the software.]

(9 Marks)

$$z \text{ Confidence Intervals for Means : } \bar{X} - \frac{z \cdot \sigma}{\sqrt{n}} < \mu < \bar{X} + \frac{z \cdot \sigma}{\sqrt{n}}$$

$z_{\alpha/2}$ values for three common confidence intervals:

Confidence Level	z
90%	1.65
95%	1.96
99%	2.58

The formula :

$$\bar{X} - \frac{z^* \sigma}{\sqrt{n}} < \mu < \bar{X} + \frac{z^* \sigma}{\sqrt{n}}$$

$\bar{X} = \text{mean} = 29.46$

$\sigma = \text{Sample Standard Deviation} = 71.27$

$n = \text{Sample size} = 30$

For 90 %:

$$29.46 - \frac{1.65 * 71.27}{\sqrt{30}} < \mu < 29.46 + \frac{1.65 * 71.27}{\sqrt{30}}$$

$$7.99 < \mu < 50.93$$

The 90% interval of estimate contains the population mean is: (7.99, 50.93)

For 95%:

$$29.46 - \frac{1.96 * 71.27}{\sqrt{30}} < \mu < 29.46 + \frac{1.96 * 71.27}{\sqrt{30}}$$

$$3.96 < \mu < 54.96$$

The 95% interval of estimate contains the population mean is: (3.96, 54.96)

For 99%:

$$29.46 - \frac{2.58 * 71.27}{\sqrt{30}} < \mu < 29.46 + \frac{2.58 * 71.27}{\sqrt{30}}$$

$$-4.11 < \mu < 63.03$$

The 99% interval of estimate contains the population mean is: (-4.11, 63.03)

- c) Comment on the values of the confidence intervals of the population mean. Why it is important to calculate confidence interval in addition to the point-estimate for the population mean?

(3 Marks)

The confidence interval for the population mean show the range in which we expect the average to fall. Depending on how confident we want to be (90% , 95% or 99%)

- d) Calculate the widths of the confidence intervals for the confidence levels given. Does the confidence interval width increase or decrease when the confidence level increases? Justify your answer

(3 Marks)

widths of the confidence intervals = upper bound – lower bound

1. **90% Confidence Interval:** Width = 50.93 - 7.99 = 42.94
2. **95% Confidence Interval:** Width = 54.96 - 3.96 = 51.0
3. **99% Confidence Interval:** Width = 63.03 - (-4.11) = 67.14

As we can see, the confidence interval width **increases as the confidence level increases**. This is because a higher confidence level requires a wider interval to be more certain that the true population mean lies within it. In other words, to be more confident, we need to cast a wider net.

-----END OF SECTION-----

SECTION IV: Hypothesis testing [CLO4]

(17 marks)

- a) Write down a clear '**Research Problem**' on your chosen data set to conduct a hypothesis test for single sample means. This research problem should be a claim on the chosen data set that you wish to either verify or dispute using a suitable hypothesis test. (3 Marks)

After performing statistical calculations, we determined an average number of followers to be **29.46 million** with a standard deviation of **71.27 million**. Our teachers hypothesize that the true average number of followers is **5 million**. To test this hypothesis, we will conduct a hypothesis test at the **10% significance level** with **sample of 30** to determine whether we should reject the null hypothesis.

- b) Perform Hypothesis Testing for the research problem. Choose any confidence/significance level, type of test, and any other data you need to conduct the hypothesis test. Show all your work clearly, including calculations, the sketch of the normal curve, the chosen test, and the values of the level of significance, critical value, etc. (9 Marks)

Here is the information we need:

Null Hypothesis (H_0): $\mu \neq 5$

Alternative Hypothesis (H_1): $\mu = 5$

Test Type: two-tailed test.

Sample Mean: 5

Assumed Population Mean: 29.46

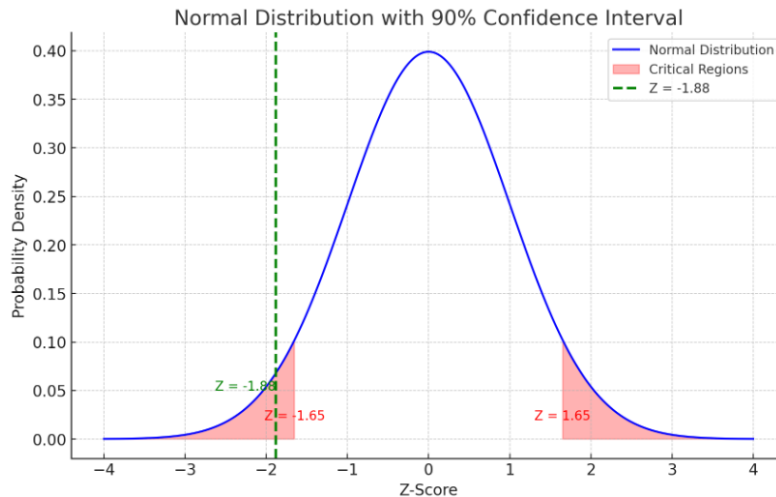
Assumed Population Standard Deviation: 71.27

Z-value for 90% confidence with one tail: ± 1.65

Sample Size : 30

$$Z = (5 - 29.46) / (71.27 / \sqrt{30})$$

$$Z = -1.88$$



- c) Summarize the results (What do the test results indicate – Reject the Null hypothesis OR Fail to reject the Null Hypothesis?). Write a Final Conclusion connecting your results obtained above with the stated “Research Problem” (5 Marks)

Based on the graph above, the calculated Z-score of **-1.88** falls within the rejection region (beyond the critical value of **-1.65**) and lies outside the range of acceptance. Therefore, we reject the null hypothesis.

This indicates that the hypothesized average number of followers, **5 million**, is not accurate.

-----END OF SECTION-----

MILESTONES

Milestones and Feedback Overview for All Deliverables:

Milestone 1: Data Collection and Descriptive Statistics (Week 10)

- Deliverable:
 - Submit the research problem, data collection methodology, and initial descriptive statistics (e.g., measures of central tendency, histograms, and boxplots).
- Feedback:
 - Instructor provides constructive feedback by:
 - Acknowledging the progress made
 - Highlighting the strengths of the work
 - Suggesting areas for improvement

Milestone 2: Hypothesis Testing and Normality Check (Week 12)

- Deliverable:
 - Submit hypothesis formulation, testing results, and normality check (including Pearson’s coefficient and outlier detection).
- Feedback:
 - Instructor provides constructive feedback by:

- Acknowledging the progress made
- Highlighting the strengths of the work
- Suggesting areas for improvement

Milestone 3: Final Report and Oral Defense (Week 14)

- Deliverable:
 - Submit the final project report, which includes data collection, descriptive statistics, hypothesis testing, and normality check. Prepare and conduct the oral defense.
- Feedback:
 - Faculty provide comprehensive feedback on the final report's structure and content, as well as the student's performance in the oral defense, assessing their understanding and articulation of the project findings.

Project Report Format:

Include the cover page, page numbers, table of contents, and/or appendices in the MS Word report.

PROJECT DELIVERABLES

1. **Report [50 Marks]:**
 - a. SECTION I: Descriptive Statistics and Sampling [CLOs 1, 3, 4] [12 marks]
 - b. SECTION II: Normality check [CLO3] [05 marks]
 - c. SECTION III: Estimation of Population Mean [CLO 4] [16 marks]
 - d. SECTION IV: Hypothesis Testing [CLO4] [17 marks]
2. **Oral Defense and Follow-up Questions and Discussion:**

Report walk-through and a PowerPoint presentation [All CLOs] [50 Marks]

Point 1 contributes to group grading [50%] and Point 2 contributes to individual grading.

Individual Grading Rubric (50%) for Oral Defense

CLO	Marks	Competencies Assessed
CLO1: Descriptive Statistics	8 Marks	Competency in summarizing data using descriptive statistics (mean, median, mode, standard deviation) and producing visualizations (e.g., histograms, boxplots).
CLO3: Sampling and Probability	8 Marks	Proficiency in applying sampling methods and computing probabilities related to continuous data distributions.
CLO4: Inferential Statistical Methods	34 Marks	Demonstrates mastery in applying inferential statistics (estimation techniques, hypothesis testing) to generalize from sample data and draw conclusions.

RUBRIC

		Absent (0) (F)	Insufficient (1-59.49%) (F)	Emerging (60-69.49%) (D/D+/C-)	Satisfactory (70-76.49%) (C/C+)	Competent (77-86.49%) (B-/B/B+)	Mastering (87-100%) (A-/A)
Group Grading [50%]	CLO1 Deliverables: Descriptive Statistics- Measures of central tendency and dispersion		1 to 4.8 points 2 or more of the following • Less than 3 correct values of sample statistics • None of the values are rounded appropriately. • Histogram and/or boxplot missing or incorrect • Comment on the shape of histogram missing	4.9 to 5.5 points Less than 2 of the following • 4 to 5 correct values of sample statistics • Only a few values are rounded appropriately • Histogram and/or boxplot missing or incorrect • Comment on the shape of histogram missing	5.6 to 6.1 points Any 2 of the following • 6 to 7 correct values of sample statistics. • Some values are rounded appropriately • Histogram and/or boxplot missing or incorrect • Comment on the shape of histogram missing	6.2 to 6.9 points Only one of the following • 8 to 9 correct values of sample statistics. • Most values are rounded appropriately • Histogram and/or boxplot missing or incorrect • Comment on the shape of histogram missing	7 to 8 points All the following • 9 to 10 correct values of sample statistics. • All values are rounded appropriately • Histogram and/or boxplot missing or incorrect • Comment on the shape of histogram missing
	CLO3 Deliverable : Sampling [4] + Normality check[6] Total 10 Marks		1 to 4.8 points 2 or more of the following • Sampling methods not described clearly • Normality calculations incomplete or partially correct • Normality comments missing or partial	4.9 to 5.5 points Less than 2 of the following • Sampling methods not described clearly • Some errors in calculations for Pearson's Coefficient or outliers • Incomplete comment on Normality	5.6 to 6.1 points 2 of the following • Sampling methods described correctly • No errors in calculations for Pearson's Coefficient or outliers • Complete and correct comment on Normality	6.2 to 6.9 points All of the following • Sampling methods described correctly • No errors in calculations for Pearson's Coefficient or outliers • Complete and correct comment on Normality	7 to 8 points All the following are true. • Sampling methods described clearly • All calculations for Pearson's Coefficient or outliers are correct • Normality comment correctly stated.

		1 to 20.4 points	20.5 to 23.4 points	23.5 to 25.9 points	26.0 to 29.3 Points	29.4 to 34 points
		Less than 2 of the following	2 of the following	3 of the following	4 of the followings	All of the followings
	CLO4 Deliverable Estimation[12] + Hypothesis Testing [13]	• All calculations for estimation are correct and complete • Inferences on confidence intervals are presented • changes in confidence interval widths are discussed • Hypothesis research problem is stated well. • Conclusion missing and/or incorrect.	• All calculations for estimation are correct and complete • Inferences on confidence intervals are presented • changes in confidence interval widths are discussed • Hypothesis research problem is stated well. • Conclusion missing and/or incorrect.	• All calculations for estimation are correct and complete • Inferences on confidence intervals are presented • changes in confidence interval widths are discussed • Hypothesis research problem is stated well. • Conclusion missing and/or incorrect.	• All calculations for estimation are correct and complete • Inferences on confidence intervals are presented • changes in confidence interval widths are discussed • Hypothesis research problem is stated well. • Conclusion missing and/or incorrect.	• All calculations for estimation are correct and complete • Inferences on confidence intervals are presented • changes in confidence interval widths are discussed • Hypothesis research problem is stated well. • Conclusion missing and/or incorrect.

Individual Grading [50%]	Oral Defense and Follow-up Questions and Discussion [50%] CLO1: 8 Mark - CLO2: 1 Mark - CLO3: 8 Marks - CLO4: 33 Marks	1 to 29.5 points	30 to 34.5 points	35 to 38 points	38.5 to 43 points	43.5 to 50 points
		Less than two of the following: - Communicates information and ideas with clarity. - Uses language with accuracy and effectiveness. - Use of appropriate IT tools to present ideas. - Able to answer all questions from the examining board.	Two of the following: - Communicates information and ideas with clarity. - Uses language with accuracy and effectiveness. - Use of appropriate IT tools to present ideas. - Able to answer all questions from the examining board.	Three of the following: - Communicates information and ideas with clarity. - Uses language with accuracy and effectiveness. - Use of appropriate IT tools to present ideas. - Able to answer all questions from the examining board.	All the following: - Communicates information and ideas with clarity. - Uses language with accuracy and effectiveness. - Use of appropriate IT tools to present ideas. - Able to answer all questions from the examining board.	All the following: - Communicates information and ideas with full clarity - Uses language with accuracy and effectiveness. - Student displays relaxed, self-confident, with no mistakes. - Use of appropriate IT tools to present ideas. - Able to answer all questions and demonstrate a complete understanding of the study.